A Measure-Theoretic Model of Accounting with Probabilistic and Temporal Frameworks

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Abstract

We develop a rigorous measure-theoretic foundation for analytical accounting. By defining an accounting space over an accounting entity as a measurable set of elementary accounts within the entity equipped with a signed measure representing the balance, we establish a static mathematical space for accounting analysis. By imposing axioms of countable additivity and zero total balance, we derive the classical accounting equation as a theorem via the Hahn–Jordan decomposition. We express accounting attributes, such as risk, profitability, and maturity, as measurable functions on the accounting spaces and explore the interpretations of integrating them with respect to signed measures, enabling the quantitative analysis of subaccounts. The total variation measure of the balance induces a natural probability space from an accounting space, where each account is weighted by its absolute value. We then explore value-weighted expectations, conditional expectations of attributes and link them with accounting interpretation and examples. We also develop families of probability measures such as power-weighted, and exponentially tilted (Esscher) measures, which are suitable for specific analytical tasks. Power-weighted measures and Rényi entropy provide tools for concentration stress testing; the Esscher measure links to regulatory capital via the VaR-based Probability Equivalent Level. The framework extends to temporal settings, capturing càdlàg balance paths, transactions as jumps, and time-aggregated risk measures essential for standards such as CECL. Concrete examples drawn from credit card portfolio dynamics, loan concentration analysis, and lifetime expected loss calculation demonstrate the practical utility of the model. The paper establishes a formal bridge between accounting and probability theory, offering a mathematically sound basis for quantitative risk management and forward-looking financial analysis.

Keywords: Accounting theory, Measure theory, Probability theory, Induced probability, Esscher transform, Rényi entropy, Temporal accounting, Risk analysis

1 Introduction

Double-entry accounting is synonymous with conventional accounting, yet its mathematical foundations have remained historically underdeveloped. There had been numerous attempts to formalise accounting, but recent attempts have focused on measurement of physical objects in monetary terms. In this regard, Ijiri [1] formulated several axioms from which one should be able to derive a monetary value of a physical object by using those axioms. In general, contributions in this direction aim to approximate the conventional accounting system rather than investigate the intrinsic properties of the accounting structure [2].

Another distinct approach can be exemplified by the structuralist reconstruction of Balzer and Mattessich [3], which formulates accounting as an empirical theory with core model and a net of specialisations. Their work provides a set theoretic axiomatisation of accounts, double-classification, and the representation of economic transactions. They also proved that the accounting equilibrium follows from their axioms, but they also treated the measurement problem, assigning monetary value to economic objects, as a central empirical issue. Their framework is focused on the representation of economic events and emphasises the empirical status of accounting.

In contrast to the above approaches, this paper takes a different perspective. Instead of treating economic transactions or valuation as the foundation, the paper instead focuses on the structure of accounts themselves. Using Kolmogorov's axiomatisation of probability theory as a reference, we use measure theory to represent accounting entities as a measurable space of elementary accounts equipped with the signed measure called balance. The axioms imposed on the accounting space are minimal: countable additivity, empty set condition, and the zero total balance. From these, we show that the accounting equation emerges as a result of the definition of the corporate entity and Hahn-Jordan decomposition. Moreover, this measure-theoretic foundation allows us to define attributes, which are measurable functions on accounting spaces, which formalises the characteristics of accounts often used in conventional accounting analysis such as risk, loss, profitability, etc.

A key part of this paper is the construction of different probability spaces from the balance function of the accounting space. The paper introduces several probability measures: starting with the standard induced probability measure, constructed from the total variation measure of the balance, power-weighted measures for concentration analysis and stress testing, and exponentially tilted (Esscher) measures to link pricing and regulatory capital. Finally, the paper extends the static framework to a temporal framework, enabling dynamic analysis of credit card portfolios, interest income, and other accounting-related problems, whilst making it suitable for the accounting standards such as the CECL.

The paper is organised as follows; in section 2, we develop the definitions of elementary accounts, accounts, balances, accounting space, and attributes of accounts and explore the integration with respect to the accounting space measure. In section 3, we construct the standard induced probability space from an accounting space by considering the total variation measure.In section 4, the paper defines more specialised probability measures for risk analysis, such as power-weighted probabilities

and Esscher Measures, and formalises concentration analysis and regulatory capital interpretation. In section 5, the paper extends the static accounting space to a temporal accounting space, including transactions, temporal mixture measures, and demonstrates their usage to standardise accounting analysis. Section 6 concludes and discusses directions for future research.

2 Static Accounting Space

In this section, we begin by constructing a static accounting space, a mathematical space to conduct an analysis of accounts at a specific point in time. Our approach is inspired by the formalisation of Probability Theory, so we used Komolgorov-style axioms to develop the structure of accounts from first principles.

2.1 Mathematical Construction of Accounts

To motivate our construction, we discuss a property of accounts that we implicitly assume in conventional accounting. Consider an account titled "Account-Receivables". In practice, an accountant often subdivides this account into smaller subaccounts (by customer or region) to maintain accuracy. The total balance of the account is the sum of the balances of its subaccounts, and these subaccounts may themselves be further subdivided (see the figure 2.1 for this hierarchical decomposition); we claim that extending this subdivision process indefinitely leads to the notion of a *elementary account*, an account that may not be subdivided any further.

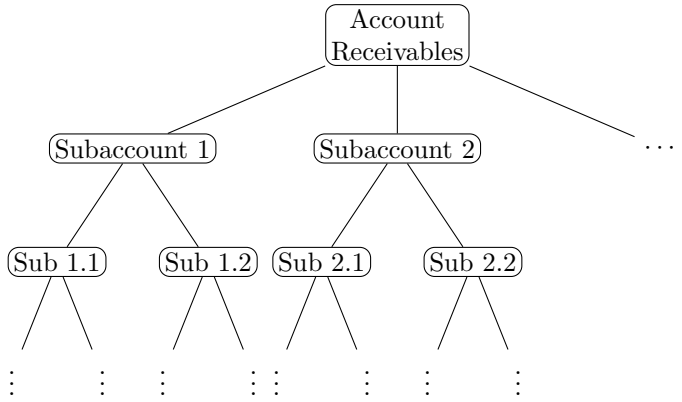


Fig. 1 Hierarchical splitting of an account into subaccounts. The process can be continued indefinitely, eventually yielding "elementary accounts" at the limit.

The existence of such elementary accounts is not heuristic; it can be established rigorously from a standard result in real analysis. In conventional accounting, an account is tied to a real number, which we call an account's balance, and subdividing the account corresponds to partitioning that real number into components. The nested interval property of closed intervals already hints at the feasibility of this partitioning:

Lemma 1 (Nested Interval Property for Closed Intervals) *For any real closed interval $I \subset \mathbb{R}$, let the sequence of closed bounded nested intervals be $I_n = [a_n, b_n]$, where $n \in \mathbb{N}$. Then, there exists a $\zeta \in \mathbb{R}$, such that $\zeta \in I_n$ for all $n \in \mathbb{N}$. In other words,*

$$\bigcap_{i=1}^{\infty} I_n \neq \emptyset$$

The next theorem strengthens this result by constructing, for each point in an interval, a canonical nested sequence whose intersection is exactly that point. This provides a mathematical representation of an elementary account.

Theorem 2.1 (Account Construction Theorem) *Let $I = [a, b] \subset \mathbb{R}$ with $a \leq b$. For each $k \in [a, b]$ and each $n \in \mathbb{N}$ (starting at $n = 1$), define the closed interval*

$$I_n(k) = \left[\max\{a,\ k - \tfrac{b-a}{2^n}\},\ \min\{b,\ k + \tfrac{b-a}{2^n}\} \right].$$

Then:

1. *Each $I_n(k)$ is a closed subinterval of $[a, b]$.*
2. *The sequence $(I_n(k))_{n=1}^{\infty}$ is nested: $I_{n+1}(k) \subseteq I_n(k)$ for all n.*
3. *The intersection over all n is the singleton $\{k\}$:*

$$\bigcap_{n=1}^{\infty} I_n(k) = \{k\}.$$

4. *The mapping $k \mapsto (I_n(k))_{n=1}^{\infty}$ is injective, hence a bijection between $[a, b]$ and the family $\{(I_n(k))_n \mid k \in [a, b]\}$.*

Proof Provided in Appendix A. □

Remark 1 Theorem 2.1 establishes a bijection between points of an interval and a family of nested interval sequences. Each point k corresponds to an elementary account that cannot be further subdivided, as its associated nested sequence shrinks to a singleton. In conventional accounting, the balance of an account is a single number; subdividing the account partitions that number into components, and the limit of this process yields elementary accounts whose balances are, in the continuous idealisation, infinitesimal. This justifies viewing an account as a continuum of elementary accounts, each bearing an indivisible unit of value.

We may now formally define an account.

Definition 2.1 (Account) Let $I = [0, \tau] \subset \mathbb{R}$ or $[\tau, 0] \subset \mathbb{R}$, where $\tau \in \mathbb{R}$, and let X be the set $\{f_k : k \in I\}$ following from theorem 2.1. We call X an account if there exists a bijection between X and I, and we call each element $f_k \in X$ an elementary account.

Remark 2 If $\tau \leq 0$, the interval is flipped in the other direction, $I = [\tau, 0]$, indicating a negative balance.

Theorem 2.1 guarantees the existence of the bijection in definition 2.1; thus an account can be viewed as a continuum of elementary accounts. This permits arbitrarily fine subdivisions of the account and its balance. Although the account itself has been defined, the definition of a balance of an account should consider the subtleties; an account may contain contraccounts whose balances have opposite signs. Consequently, subdividing a positive balance can yield subaccounts with negative balances. This motivates the following definition:

Definition 2.2 (Balance) Define a measurable (X, Σ) space to be an account space, where X is an account and Σ is a $\sigma-$ algebra defined on X. Let X be mapped to $I = [0, \tau]$, which is the interval as defined previously. Then we call the signed measurable function $\text{bal} : \Sigma \to \mathbb{R}$ the function which satisfies the following:

Axiom 1 (Empty Set)

$$\text{bal}(\emptyset) = 0$$

Axiom 2 (Balance of an Account)

$$\text{bal}(X) = \tau$$

Axiom 3 (Additivity) *For any countable collection of disjoint sets* $\{X_i\} \subset \Sigma$*, we have*

$$\text{bal}\,(\bigcup_{i=1}^{\infty} X_i) = \sum_{i=1}^{\infty} \text{bal}(X_i)$$

Naturally, we may define a subaccount from the definition of an account.

Definition 2.3 (Subaccount) If X is an account, then S is a subaccount of X if $S \subset X$.

Here, we note that the measure theoretic construction of accounts and balances yield several standard properties. The following theorem states the most essential properties.

Theorem 2.2 (Basic Properties of Balance) *Let* (X, Σ, bal) *be an account space. Then:*

1. *Complement: For any subaccount* $S \subset X$*,*

$$\text{bal}(S^c) = \text{bal}(X) - \text{bal}(S).$$

2. *Monotonicity: If* $A \subset B$ *and* bal *is a positive measure, then* $\text{bal}(A) \leq \text{bal}(B)$*. (For signed measures, this holds for the total variation measure* $|\,\text{bal}\,|$*.)*
3. *Continuity from below: If* $\{S_j\}_{j \in \mathbb{N}}$ *is an increasing sequence of subaccounts* ($S_j \subset S_{j+1}$)*, then*

$$\text{bal}\Big(\bigcup_{j=1}^{\infty} S_j\Big) = \lim_{j \to \infty} \text{bal}(S_j).$$

4. *Continuity from above: If* $\{S_j\}_{j\in\mathbb{N}}$ *is a decreasing sequence* ($S_j \supset S_{j+1}$) *and* $\operatorname{bal}(S_1)$ *is finite, then*

$$\operatorname{bal}\Big(\bigcap_{j=1}^{\infty} S_j\Big) = \lim_{j\to\infty} \operatorname{bal}(S_j).$$

5. *Finite additivity: For any finite collection of disjoint subaccounts* $S_1, \ldots, S_n$,

$$\operatorname{bal}\Big(\bigcup_{i=1}^{n} S_i\Big) = \sum_{i=1}^{n} \operatorname{bal}(S_i).$$

6. *Countable subadditivity: For any countable collection of subaccounts* $\{S_j\}_{j\in\mathbb{N}}$,

$$\operatorname{bal}\Big(\bigcup_{j=1}^{\infty} S_j\Big) \leq \sum_{j=1}^{\infty} \operatorname{bal}(S_j),$$

where the inequality holds for positive measures; for signed measures it applies to $|\operatorname{bal}|$.

2.2 Derivation of Basic Accounting Principles

With the formal definition of the account established, we now attempt to formalise the notion of an *entity*. In conventional accounting, an entity is often defined to be a collection of accounts. We adopt this perspective.

Definition 2.4 (Entity) An accounting entity is a set whose elements are accounts.

We now introduce the central object in our framework: accounting space. This space provides the mathematical basis to analyse a financial position of an entity.

Definition 2.5 An Accounting Space of an entity is a triplet (γ, λ, T), where γ is the set of all elementary accounts belonging to the entity, λ is $\sigma-$algebra on γ, and $T : \lambda \to \mathbb{R}$ is a signed measure satisfying the following:

Axiom 4 (Additivity) *For any countable pairwise disjoint sets* $\{X_i\}_{i=1}^{\infty} \subset \lambda$,

$$T(\bigcup_{i=1}^{\infty} X_i) = \sum_{i=1}^{\infty} T(X_i)$$

Axiom 5 (Balance)

$$T(\gamma) = 0$$

Axiom 6 (Empty Account)

$$T(\emptyset) = 0$$

Remark 3 Axiom 5 is the major additional axiom in an accounting space.

We note a simple consequence of the axioms before turning to decomposition theorems.

Proposition 2.3 *Let (γ, λ, T) be an accounting space and let $\{X_i\}_{i\in I}$ be a countable partition of γ into measurable sets. Then it is impossible that $T(X_i) > 0$ for all i, and impossible that $T(X_i) < 0$ for all i. In particular, if T is not identically zero, there exist indices i, j such that $T(X_i) > 0$ and $T(X_j) < 0$.*

Proof If $T(X_i) > 0$ for all i, then by countable additivity $T(\gamma) = \sum_i T(X_i) > 0$, contradicting Axiom 5. The case where all $T(X_i)$ are negative follows similarly. □

Proposition 2.3 shows that any non-trivial accounting space must contain both positive and negative balances. This motivates the existence of a more refined decomposition, and because an accounting space is a measurable space equipped with a signed measure, the classical decomposition theorems in measure theory apply directly.

Theorem 2.4 (Hahn Decomposition Theorem for Accounting Spaces) *Let (γ, λ, T) be an accounting space. Then there exists a measurable partition $\gamma = P \cup N$ with $P \cap N = \emptyset$ such that:*

1. *For every measurable set $A \subseteq P$, we have $T(A) \geq 0$.*
2. *For every measurable set $A \subseteq N$, we have $T(A) \leq 0$.*

*The pair (P, N) is called a **Hahn decomposition** of γ with respect to T. Moreover, if (P', N') is another such decomposition, then the symmetric differences $P \triangle P'$ and $N \triangle N'$ are T-null sets.*

Proof This is the classical Hahn decomposition theorem applied to the signed measure T on the measurable space (γ, λ). □

The Hahn decomposition separates the entity's elementary accounts into those with non-negative balance contributions and those with non-positive balance contributions. Building on this, we obtain the Jordan decomposition, which resolves a signed measure into the difference of two positive measures.

Theorem 2.5 (Jordan Decomposition Theorem for Accounting Spaces) *Let (γ, λ, T) be an accounting space and let $\gamma = P \cup N$ be a Hahn decomposition as in Theorem 2.4. Define for any $X \in \lambda$*

$$T^+(X) = T(X \cap P), \qquad T^-(X) = -T(X \cap N).$$

Then:

1. *T^+ and T^- are positive measures on (γ, λ).*
2. *$T = T^+ - T^-$.*

3. *The measures T^+ and T^- are mutually singular (concentrated on P and N, respectively).*
4. *This decomposition is unique: if $T = \mu - \nu$ with μ, ν positive and mutually singular, then $\mu = T^+$ and $\nu = T^-$.*

Proof This is the classical Jordan decomposition theorem applied to the signed measure T. Positivity and mutual singularity follow directly from the construction using the Hahn decomposition. Uniqueness is a standard result. □

In the accounting context, we call T^+ the debit measure and T^- the credit measure. We now define the conventional corporate accounting entity.

Definition 2.6 (Corporate Entity) A **corporate entity** is an accounting entity whose accounts are partitioned into three disjoint subaccounts: assets A, liabilities L, and equity S. For its accounting space (γ, λ, T), we require that the Hahn decomposition of Theorem 2.4 satisfies $P = A$ and $N = L \cup S$; that is, assets constitute the positive set, while liabilities and equity together constitute the negative set.

The above definition is precisely how a corporate entity is defined in conventional accounting, where asset accounts are usually measured with debits, and liabilities and equity accounts are usually measured with credits. From this definition, the fundamental equation of the accounting emerges directly.

Theorem 2.6 (Accounting Equation) *For any corporate entity as defined in Definition 2.6,*

$$T^+(A) = T^-(L) + T^-(S)$$

Proof From Axiom 5 we have

$$T(A) + T(L) + T(S) = 0,$$

hence

$$T(A) = -\big(T(L) + T(S)\big).$$

By the Jordan decomposition (Theorem 2.5), $T(A) = T^+(A)$ because $A \subseteq P$. Similarly, $T(L) = -T^-(L)$ and $T(S) = -T^-(S)$ because $L, S \subseteq N$. Substituting these into the previous equality yields

$$T^+(A) = T^-(L) + T^-(S)$$

□

Thus, the fundamental equation of double-entry accounting is not an assumption but a theorem derived from the axioms of an accounting space along with the definition of a corporate entity.

2.3 Attributes of Accounts and Integration

In conventional accounting, we usually attach quantitative characteristics to accounts; for example, the loss given default, the maturity, risk rating, etc. These characteristics, which we call *attributes* are essential to risk analysis, performance measurement, and regulatory reporting. In order to incorporate the notion of attributes into our measure theoretic model, we define them as the following:

Definition 2.7 (Attribute) Let $\mathcal{A} = (\gamma, \lambda, T)$ be an accounting space. An attribute on $\mathcal{A}$ is any measurable function $f : \gamma \to \mathbb{R}$.

An attribute f would assign each elementary account $x \in \gamma$ a real number $f(x)$, which would represent some quantity of interest. We may define an attribute over a subaccount as the following:

Definition 2.8 (Attribute of an Account) Let $\mathcal{A} = (\gamma, \lambda, T)$ be an accounting space, and f an attribute on $\mathcal{A}$. For any subaccount $A \in \lambda$, the attribute of A is defined as

$$\int_A f d|T| \coloneqq \int_A f dT^+ - \int_A f dT^-$$

However, one may wish to integrate without regard for the sign, so in that case, we may use the total variation measure $|T| \coloneqq T^+ + T^-$ to define the following:

Definition 2.9 (Total Attribute of an Account) Let $\mathcal{A} = (\gamma, \lambda, T)$ be an accounting space, and f an attribute on $\mathcal{A}$. For any subaccount $A \in \lambda$, the total attribute of A is defined as

$$\int_A f d|T| \coloneqq \int_A f dT^+ + \int_A f dT^-$$

We may use the above integrals to standardise problems related to attributes of accounts. For example,

Example 1 (Loss Given Default) Let $\ell : \gamma \to [0, 1]$ be the loss given default associated with each account. For a subaccount A (e.g., all loans in a certain sector), the total exposure weighted by loss given default is

$$\int_A \ell \, d|T| = \int_A \ell \, dT^+ + \int_A \ell \, dT^-.$$

Because loss given default is a positive quantity, we integrate with respect to $|T|$ to ensure that both assets and liabilities contribute positively to the total exposure measure.

Example 2 (Indicator of a Condition) Let χ_A be the indicator attribute of a subaccount A itself, i.e., $\chi_A(x) = 1$ if $x \in A$ and 0 otherwise. Then

$$\int_\gamma \chi_A \, dT = T(A),$$

the balance of A. Similarly, $\int_\gamma \chi_A \, d|T| = |T|(A)$, the total absolute value in A.

Example 3 (Geographic Concentration) Suppose γ is partitioned into regions $R_1, \ldots, R_k$ and let $g_i(x) = 1$ if $x \in R_i$, 0 otherwise. The total absolute value in region i is $\int_\gamma g_i \, d|T| = |T|(R_i)$. This quantity can be compared across regions to assess geographic concentration.

Example 4 (Maturity-Weighted Balance) Let $t : \gamma \to [0, \infty)$ be the remaining maturity (in years) of each account. Then the maturity-weighted balance of a subaccount A is $\int_A t \, d|T|$. This measure is used in interest rate risk analysis, where longer maturities imply greater sensitivity to rate changes.

Moreover, the following properties follow from the definition of the integral and the axioms of a signed measure.

Proposition 2.7 (Basic Properties of Total Attribute) *Let $\mathcal{A} = (\gamma, \lambda, T)$ be an accounting space, let f, g be attributes, and let $A, B \in \lambda$ be disjoint subaccounts. Then:*

1. *Linearity: For any real constants α, β,*

$$\int_A (\alpha f + \beta g) \, dT = \alpha \int_A f \, dT + \beta \int_A g \, dT,$$

   and similarly for $|T|$.
2. *Additivity over disjoint sets:*

$$\int_{A \cup B} f \, dT = \int_A f \, dT + \int_B f \, dT.$$

3. *Monotonicity (for $|T|$): If $f \leq g$ pointwise, then*

$$\int_A f \, d|T| \leq \int_A g \, d|T|.$$

4. *Triangle inequality:*

$$\left| \int_A f \, dT \right| \leq \int_A |f| \, d|T|.$$

5. *Constant attribute: If $f(x) = c$ for all x, then $\int_A c \, dT = c\,T(A)$ and $\int_A c \, d|T| = c\,|T|(A)$.*
6. *Convergence: If $\{f_n\}$ is a sequence of attributes converging pointwise to f and g is an integrable dominating function for $|T|$ (i.e., $|f_n| \leq g$ and $\int g \, d|T| < \infty$), then*

$$\lim_{n \to \infty} \int_A f_n \, dT = \int_A f \, dT$$

   and similarly for $|T|$.

Proof All statements follow from the corresponding properties of the Lebesgue integral applied to T^+ and T^- separately, together with the definition of the signed integral. The convergence theorems are standard results from measure theory applied to the positive measure $|T|$ and then extended to T via its decomposition. □

The above properties ensure that attributes behave predictably under aggregation and limits, essential for applications involving large portfolios, continuous-time models, or asymptotic approximations. The framework of attributes will be used extensively in the following sections.

3 Induced Probability Space

In this section, we examine the construction of an induced probability space from an accounting space. We shall assume that the accounting entities contain a countable number of accounts for simplicity, though the construction extends to the general case.

3.1 Construction from the Total Variation Measure

An accounting space consists of accounts that are debit and credit measures, corresponding to positive and negative signs respectively. To obtain a non-negative measure for a probability space, we use the total variation measure, which assigns the total absolute value to each subaccount. We denote the total mass of the entity by $\|T\| \coloneqq |T|(\gamma)$. If we assume $\|T\| > 0$, we may normalise $|T|$ to construct a probability measure:

Definition 3.1 (Standard Induced Probability) Let (γ, λ, T) be an accounting entity with $\|T\| > 0$, then we may define a probability measure $\mathbb{P} : \gamma \to [0, 1]$ by

$$\mathbb{P}(E) = \frac{|T|(E)}{\|T\|}, \forall E \in \lambda$$

We call (γ, λ, T) the induced probability space of the accounting entity, and we also call $\mathbb{P}$ the standard probability measure for an accounting space.

For any subaccount $E \in \lambda$, $\mathbb{P}(E)$ represents the proportion of the entity's total absolute balance. Thus $\mathbb{P}$ is a value-weighted distribution, which is similar to the "sized-based" sampling measure. We may then proceed to find basic properties of $\mathbb{P}$.

Theorem 3.1 (Balance between positive and negative sets) *Let (P, N) Hahn-Jordan Decomposition of an accounting entity (γ, λ, T), where P, N represents positive and negative sets respectively. If $(\gamma, \lambda, \mathbb{P})$ is the induced probability space, then*

$$\mathbb{P}(N) = \mathbb{P}(P) = \frac{1}{2}$$

Proof Note that we have $\mathbb{P}(N) = \mathbb{P}(P)$, since $T^+(P) = T^-(N)$ and $T(\gamma) = T^+(N) - T^-(P) = 0$ by assumption of the accounting space. Moreover, we have $\mathbb{P}(\gamma) = 1$, so it follows that $\mathbb{P}(N) + \mathbb{P}(P) = 1$ and it immediately follows that $\mathbb{P}(N) = \mathbb{P}(P) = \frac{1}{2}$ □

For a corporate entity, the positive set constitutes the assets, whereas the negative set constitutes liabilities and equity, yielding the following corollary.

Corollary 3.2 *Let (γ, λ, T) represent a corporate entity with decomposition $(A, L \cup S)$, where A, L, S represents assets, liabilities, stockholders' equity accounts respectively, then we have*

$$\mathbb{P}(A) = \mathbb{P}(L \cup S) = \frac{1}{2}$$

This Corollary 3.2 simply formalises the accounting principle that asset and liability side carry equal weight.

3.2 Expectation

Having established an induced probability space, we naturally interpret attributes as random variables to define their expectations.

Definition 3.2 (Expected Value of an attribute) Let $\mathcal{A} = (\gamma, \lambda, T)$ be an accounting space equipped with the standard probability measure $\mathbb{P}$, then for any attribute f on $\mathcal{A}$, the expected value of the attribute is given by

$$\mathbb{E}_{\mathbb{P}}[f] = \int f d\mathbb{P} = \frac{1}{\|T\|} \int f d|T|$$

Remark 4 $\mathbb{E}_{\mathbb{P}}[f]$ is precisely the value-weighted average of the attribute across the underlying entity of the accounting space. Note also that under the measure $\mathbb{P}$, the attribute f can be interpreted as a random variable with a distribution $P \circ f^{-1}$.

Example 5 (Probability of Default) Let $p(x)$ be the one-year probability of default for account x. Then $\mathbb{E}_{\mathbb{P}}[p]$ is the average probability of default per dollar of absolute value, a standard measure of portfolio credit quality.

Since $\mathbb{P}$ weighs positive and negative accounts symmetrically, the expectation of an attribute value may not be fully captured by Definition 3.2.

Example 6 (Profit Margin) Let $m(x)$ be the profit margin per unit balance, where it is positive for profit, negative for loss. Then $\mathbb{E}_{\mathbb{P}}[m]$ is the average profit margin per dollar of absolute value. However, this quantity may not reflect net profitability, as it treats a quantity of losses (in negative balance) with the same weight as the same quantity of assets (in positive balance). For net profit, one should instead consider the signed integral $\int m\, dT$ introduced in Definition 2.9.

The symmetric weighting of $\mathbb{P}$ motivates the decomposition of the induced measure into its positive and negative parts.

Proposition 3.3 (Decomposition of Standard Probability) *Let (γ, λ, T) be an accounting space with the Jordan decomposition $T = T^+ - T^-$ and postive and negative sets P, N respectively. Let $M = T^+(\gamma) = T^-(\gamma)$, then $\mathbb{P}$ satisfies the following*

$$\mathbb{P}(E) = \frac{1}{2}\mathbb{P}^+(E \cap P) + \frac{1}{2}\mathbb{P}^-(E \cap N) \quad \forall E \in \lambda$$

where $\mathbb{P}^+(A) = T^+(A)/M$ for $A \subseteq P$, and $\mathbb{P}^-(A) = T^-(A)/M$ for $A \subseteq N$

Proof Note that by definition, we have $\mathbb{P}(A) = |T|(A)/\|T\|$, but then by definition we also have $\|T\| = T^+(\gamma) + T^-(\gamma) = 2M$, thus we have $\mathbb{P}(A) = |T|(A)/(2M)$. Here, since T^+ is concentrated on P and likewise for T^- for N, we have $T^+(E) = T(E \cap P)$ and similarly for $T^-(E) = T^-(E \cap N)$. Hence, we have

$$\mathbb{P}(E) = \frac{T^+(E \cap P)}{2M} + \frac{T^-(E \cap N)}{2M} = \frac{1}{2}\mathbb{P}^+(E \cap P) + \frac{1}{2}\mathbb{P}^-(E \cap N)$$

□

In effect, the decomposition of $\mathbb{P}$ into $\mathbb{P}^+$ and $\mathbb{P}^-$ separates the two sides of the analysis in a typical balance sheet, separating the classes between those for debit and credit measures. To show this, we shall revisit the example for the probability of default.

Example 7 (Expected loss Revisited) For a corporate entity, we have assets in P and liabilities and equities in N. Let $p(x)$ be the probability of default. In this context, the expected loss would be only relevant for assets, assuming that liabilities are obligations instead of exposures, so the expected loss per unit absolute value would be given by $\mathbb{E}_{\mathbb{P}^+}[p]$ instead of the whole $\mathbb{E}_{\mathbb{P}}[p]$

3.3 Conditional Expectation

Attributes are rarely homogeneous across an entire entity. Risk managers, for instance, must understand how risk concentrates in particular segments (geographic regions, product lines, rating classes). Conditional expectation provides the natural tool for such segment-specific analysis.

Definition 3.3 (Conditional Expectation of an Attribute) Let $(\gamma, \lambda, \mathbb{P})$ be the induced probability space of an accounting entity. Let $\mathcal{G} \subset \lambda$ be a sub-σ-algebra (representing the information by which we condition). For an integrable attribute $f : \gamma \to \mathbb{R}$, the **conditional expectation** $\mathbb{E}_{\mathbb{P}}[f \mid \mathcal{G}]$ is the unique $\mathcal{G}$-measurable function satisfying

$$\int_G \mathbb{E}_{\mathbb{P}}[f \mid \mathcal{G}]\, d\mathbb{P} = \int_G f\, d\mathbb{P} \qquad \forall\, G \in \mathcal{G}.$$

Remark 5 The definition is valid for any sub-σ-algebra, not only those generated by finite partitions. If $\mathcal{G}$ is generated by a continuous variable (e.g., account age), the conditional expectation becomes a function of that variable, describing how the attribute varies smoothly across its values.

When $\mathcal{G}$ arises from a partition, the conditional expectation takes a particularly simple form.

Proposition 3.4 *Suppose $\mathcal{G}$ is generated by a countable partition $\{E_i\}_{i\in I}$ of γ into measurable sets. Then for $\mathbb{P}$-almost every $\omega \in E_i$,*

$$\mathbb{E}_{\mathbb{P}}[f \mid \mathcal{G}](\omega) = \frac{1}{\mathbb{P}(E_i)} \int_{E_i} f \, d\mathbb{P},$$

provided $\mathbb{P}(E_i) > 0$.

Thus, on each partition element, the conditional expectation equals the value-weighted average of the attribute restricted to that element.

The most fundamental partition of an accounting entity is the separation into positive and negative accounts. Applying conditional expectation to this partition recovers the positive and negative measures introduced earlier.

Theorem 3.5 *Let $\mathcal{G}$ be the σ-algebra generated by the partition $\{P, N\}$, where P is the positive set and N the negative set. Then for any integrable attribute f,*

$$\mathbb{E}_{\mathbb{P}}[f \mid P] = \mathbb{E}_{\mathbb{P}^+}[f], \qquad \mathbb{E}_{\mathbb{P}}[f \mid N] = \mathbb{E}_{\mathbb{P}^-}[f].$$

Proof On P, Proposition 3.4 gives

$$\mathbb{E}_{\mathbb{P}}[f \mid P] = \frac{1}{\mathbb{P}(P)} \int_P f \, d\mathbb{P}.$$

Since $\mathbb{P}(P) = 1/2$ and $\int_P f \, d\mathbb{P} = \frac{1}{2} \int_P f \, d\mathbb{P}^+$ (because $d\mathbb{P} = \frac{1}{2} d\mathbb{P}^+$ on P), we obtain

$$\mathbb{E}_{\mathbb{P}}[f \mid P] = \frac{\frac{1}{2} \int_P f \, d\mathbb{P}^+}{1/2} = \int_P f \, d\mathbb{P}^+ = \mathbb{E}_{\mathbb{P}^+}[f].$$

The argument for N is symmetric. □

This theorem unifies the decomposition of $\mathbb{P}$ with the probabilistic concept of conditioning: conditioning on the asset side yields the positive measure $\mathbb{P}^+$, and conditioning on the liability-equity side yields the negative measure $\mathbb{P}^-$. It provides a rigorous foundation for analysing attributes separately on each side of the balance sheet.

4 Probability Measures for Accounting Risk Analysis

From our standard induced probability $\mathbb{P}$, we have shown that we can decompose it into $\mathbb{P}^+$ and $\mathbb{P}^-$, but in this section, we introduce a few alternative measures that can directly link our accounting space framework to known methods in risk management and analysis.

4.1 Power-Weighted Probabilities and Rényi Entropy

The standard induced probability $\mathbb{P}$ weighs each elementary account by its absolute balance, but this may not be optimal for all analytical purposes. In particular, risk managers often need to emphasise the influence of large accounts for concentration risk. Hence, we introduce a family of probability measure parametrised by $p > 0$.

Definition 4.1 (Power-Weighted Probabilities) Let (γ, λ, T) be an accounting space and let μ be a reference measure such that $|T| \ll \mu$. Denote by $h = d|T|/d\mu$ the Radon–Nikodym density of the total variation measure. For any $p > 0$, define the p-weighted measure

$$|T|_p(E) = \int_E h(x)^p \, \mu(dx), \qquad E \in \lambda,$$

and the corresponding p-weighted probability measure

$$\mathbb{P}_p(E) = \frac{|T|_p(E)}{|T|_p(\gamma)}.$$

In the discrete case where μ is counting measure, this simplifies to

$$\mathbb{P}_p(\{x\}) = \frac{|T(\{x\})|^p}{\sum_{y \in \gamma} |T(\{y\})|^p}.$$

Remark 6 The parameter p controls the sensitivity of the measure to account size:

- If $p = 1$ then it is equal to the induced probability $\mathbb{P}$.
- If $p > 1$ then the larger the value of p, the greater the concentration on larger accounts.
- If $p < 1$ then the smaller the value of p, the greater the concentration on the smaller accounts.
- As $p \to \infty$, $\mathbb{P}_p$ concentrates on the largest account(s), and if a unique maximum exists, it converges to a Dirac measure at that account.
- As $p \to 0^+$, $\mathbb{P}_p$ tends to the uniform distribution over all accounts with respect to the reference measure μ.

Power weighted probabilities can be used for stress testing. One may choose a $p > 1$ to simulate a scenario in which large exposures are magnified, revealing vulnerabilities that may not be apparent under standard weighting.

We may also accompany power-weighted probabilities with the Rényi Entropy, a measure of diversification which has already been used for analysing financial markets [4].

Definition 4.2 (Rényi Entropy) Let $(\gamma, \lambda, \mathbb{P})$ be the induced probability space of an accounting entity, with density $f = d\mathbb{P}/d\mu$ relative to a reference measure μ. For any order $p > 0$, $p \neq 1$, the Rényi entropy of order p is defined as

$$H_p(\mathbb{P}) = \frac{1}{1-p} \log \left(\int_\gamma f(x)^p \, \mu(dx) \right).$$

Remark 7 The limiting case $p \to 1$ yields the Shannon entropy

$$H_1(\mathbb{P}) = -\int_\gamma f(x) \log f(x)\, \mu(dx),$$

The case $p = 2$ is related to the Herfindahl–Hirschman Index (HHI) [5],

$$H_2(\mathbb{P}) = -\log\left(\sum_x \mathbb{P}(\{x\})^2\right) = -\log(\mathrm{HHI}),$$

where $\mathrm{HHI} = \exp(-H_2(\mathbb{P}))$.

There is a natural link between Rényi entropy and power weighted probabilities. Observe that the integral in the definition of $H_p(\mathbb{P})$ is precisely the normalising constant of $\mathbb{P}_p$:

$$\int_\gamma f(x)^p\, \mu(dx) = \frac{|T|_p(\gamma)}{\|T\|^p}.$$

Consequently,

$$|T|_p(\gamma) = \|T\|^p \exp\big((1-p)H_p(\mathbb{P})\big).$$

Definition 4.3 (Effective Number of Accounts) For an induced probability space $(\gamma, \lambda, \mathbb{P})$, the effective number of accounts of order p is defined as

$$N_p(\mathbb{P}) = \exp\big(H_p(\mathbb{P})\big).$$

Remark 8 This quantity represents the number of equal-sized accounts that would yield the same Rényi entropy as the actual distribution. Special cases include:

- $N_1 = \exp(H_1)$: the exponential of Shannon entropy.
- $N_2 = \exp(H_2) = 1/\,\mathrm{HHI}$ [5]: the inverse Herfindahl index, which can be interpreted as the number of equal sized accounts which give the same concentration.
- $N_\infty = \lim_{p\to\infty} N_p(\mathbb{P}) = 1/\max_x \mathbb{P}(\{x\})$: isolates the influence of the largest account.

Investigating how N_p varies with p can reveal the tail structure of the distribution [6]. A distribution that is relatively homogeneous tends to show N_p slowly declining as p increases. In contrast, a distribution with heavy tails, where a few but sufficiently large accounts dominate, will show a sharp drop in N_p for $p > 2$ [7]. The analysis of this behaviour may provide insights into risk management; a portfolio that appears well-diversified using N_1 may nonetheless be prone to the default of a few large obligors, which may only be revealed by N_p for larger p values.

Example 8 (Concentration Analysis of a Commercial Loan Portfolio) Let (γ, λ, T) be the accounting space of a commercial bank. The elementary accounts γ include all of the bank's assets, liabilities, and equity accounts. Consider the subaccount $A \subset \gamma$ representing the commercial loan portfolio, consisting of $n = 1000$ individual loans. Each loan $x_i \in A$ has a positive balance $b_i = T(\{x_i\}) > 0$, i.e., it is an asset. The rest of the entity (liabilities and equity) are contained in the complement $\gamma \setminus A$, with negative balances ensuring $T(\gamma) = 0$.

For concentration analysis we focus on the loan portfolio A. Its total absolute value is

$$||T_A|| := |T|(A) = \sum_{i=1}^{n} b_i,$$

and the induced probability restricted to the loans is

$$\mathbb{P}_A(\{x_i\}) = \frac{|T|(\{x_i\})}{||T_A||} = \frac{b_i}{\sum_{j=1}^{n} b_j}.$$

Now suppose the largest loan constitutes a fraction q of the total portfolio value, and the remaining 999 loans are of equal size. Then $b_{\max} = q \cdot ||T_A||$, and each other loan has balance $b = (1-q)||T_A||/999$. The effective number of accounts of order p for this portfolio (as defined in Definition 4.3) is

$$N_p(\mathbb{P}_A) = \left(q^p + 999 \left(\frac{1-q}{999} \right)^p \right)^{1/(1-p)}, \quad p \neq 1,$$

with the Shannon case $p \to 1$ giving

$$N_1(\mathbb{P}_A) = \exp\left(-q \log q - (1-q) \log \frac{1-q}{999} \right).$$

q	N_1	N_2	N_5
0.05	864.2	293.8	42.3
0.10	247.5	92.5	17.8
0.20	88.6	24.6	7.5

Interpretation within the accounting space:

- N_1 measures the diversity of the loan portfolio under the standard value-weighted distribution derived directly from the balance sheet. For $q = 0.05$, the portfolio appears as diversified as 864 equal-sized loans.
- $N_2 = 1/\,\text{HHI}$ is the inverse Herfindahl index, a standard regulatory concentration measure. At $q = 0.05$, $N_2 \approx 294$ suggests moderate concentration.
- N_5 reveals the tail structure: the portfolio behaves like only 42 equal-sized loans when higher-order concentration is emphasised. This indicates that while the portfolio appears well-diversified overall, a few large loans dominate its tail risk.

A modest increase in q from 5% to 10% reduces N_5 from 42 to 18, illustrating the sensitivity of higher-order effective numbers to tail risk. Regulators and risk managers can use such analysis to set limits on q or on N_p for selected p [6].

4.2 Esscher Measures

Whereas power-weighted measures emphasise concentration by taking the absolute balances of accounts to a power, the Esscher (Exponentially Tilted) weighs the balances differently, emphasising accounts with large values of an attribute f.

Definition 4.4 (Esscher Measure) Let (γ, λ, T) be an accounting space and let $f : \gamma \to \mathbb{R}$ be an attribute. Given a parameter $\theta \in \mathbb{R}$, the Esscher measure (or exponentially tilted measure)

of order θ is defined by

$$\mathbb{P}_\theta(E) = \frac{\displaystyle\int_E e^{\theta f(x)}\,|T|(dx)}{\displaystyle\int_\gamma e^{\theta f(x)}\,|T|(dx)}, \qquad E \in \lambda.$$

Remark 9 We also θ the loading parameter [8]. When $\theta > 0$, the measure emphasises the accounts with larger values of f; when $\theta < 0$, it emphasises accounts with smaller f. When $\theta = 0$ the measure is equal to the standard induced probability $\mathbb{P}$.

Definition 4.5 (Esscher Premium) For a given attribute f and loading parameter θ, the Esscher premium is defined as the expectation of f under the tilted measure:

$$\Pi_\theta(f) := \mathbb{E}_{\mathbb{P}_\theta}[f] = \frac{\displaystyle\int_\gamma f(x)\, e^{\theta f(x)}\,|T|(dx)}{\displaystyle\int_\gamma e^{\theta f(x)}\,|T|(dx)}.$$

Remark 10 In the context of actuarial sciences, if f represents insurance losses, then $\Pi_\theta(f)$ would be the risk-adjusted premium that an insurer should charge to cover the expected loss and the load of the risk [8].

When applying Esscher Measures in economic contexts, the choice of θ is often a difficulty, since the parameter θ is limited in interpretability. To overcome this limitation, we use the approach introduced by Wen and Zhang (2025), which links θ to a quantile-based risk measure through the concept of the VaR-based Probability Equivalent Level (PELV) [9].

To use the approach, we first interpret define VaR in our accounting space framework.

Definition 4.6 (Value at Risk) For a confidence level $\alpha \in (0, 1)$, the Value at Risk of an attribute f representing a loss of an account under $\mathbb{P}$ is

$$\mathrm{VaR}_\alpha(f) = \inf\{t \in \mathbb{R} : \mathbb{P}(f \le t) \ge \alpha\},$$

It is the smallest value such that the probability of f not exceeding it is at least α.

Now, we define the PELV as stated in Wen and Zhang (2025), adjusting it accordingly for our accounting context.

Definition 4.7 (PELV) The VaR-based Probability Equivalent Level (PELV) for an attribute f at confidence level α is

$$\delta_\alpha(f) = \sup\{\theta \ge 0 : \mathbb{E}_{\mathbb{P}_\theta}[f] \le \mathrm{VaR}_\alpha(f)\}.$$

$\delta_\alpha(f)$ is the largest loading parameter for which the Esscher premium does not exceed the VaR.

Under mild moment conditions [9], $\delta_\alpha(f)$ is the unique solution to the equation $\mathbb{E}_{\mathbb{P}_\theta}[f] = \mathrm{VaR}_\alpha(f)$, which is expressed more precisely in the following:

Theorem 4.1 (Existence and Uniqueness of the PELV) *Let $f : \gamma \to \mathbb{R}$ be a non-degenerate attribute such that the moment generating function $M(\theta) = \mathbb{E}_{\mathbb{P}}[e^{\theta f}]$ is finite for all θ in an interval $[0, \theta_{\max})$ with $\theta_{\max} > 0$ (allowing $\theta_{\max} = +\infty$). Assume that*

$$\mathbb{E}_{\mathbb{P}}[f] \leq \mathrm{VaR}_\alpha(f) < \lim_{\theta \to \theta_{\max}} \mathbb{E}_{\mathbb{P}_\theta}[f].$$

Then the PELV $\delta_\alpha(f)$ is the unique solution of $\mathbb{E}_{\mathbb{P}_\theta}[f] = \mathrm{VaR}_\alpha(f)$.

Proof The function $\varphi(\theta) = \mathbb{E}_{\mathbb{P}_\theta}[f]$ is continuous on $[0, \theta_{\max})$ and strictly increasing because

$$\varphi'(\theta) = \frac{\partial}{\partial \theta} \mathbb{E}_{\mathbb{P}_\theta}[f] = \mathrm{Var}_{\mathbb{P}_\theta}(f) > 0$$

for a non-degenerate f. By hypothesis, we have $\varphi(0) = \mathbb{E}_{\mathbb{P}}[f] \leq \mathrm{VaR}_\alpha(f)$ and $\lim_{\theta \to \theta_{\max}} \varphi(\theta) > \mathrm{VaR}_\alpha(f)$. Now, by the classical Intermediate Value Theorem, there exists a unique $\theta^* \in [0, \theta_{\max})$ with $\varphi(\theta^*) = \mathrm{VaR}_\alpha(f)$. For any $\theta < \theta^*$, by strict monotonicity, $\varphi(\theta) < \mathrm{VaR}_\alpha(f)$; for $\theta > \theta^*$, $\varphi(\theta) > \mathrm{VaR}_\alpha(f)$. Hence $\delta_\alpha(f) = \theta^*$ and $\delta_\alpha(f)$ is the unique solution of the equation. □

We claim that there is a direct link between the Esscher premium and regulatory capital of an accounting entity. This is because we use the notion of VaR when we define regulatory capital in the language of accounting spaces.

Definition 4.8 (Regulatory Capital) Let (γ, λ, T) be an accounting space of a given entity and let $\ell : \gamma \to \mathbb{R}$ be a loss attribute (expected loss, loss given default, or market decline). For a confidence level $\alpha \in (0, 1)$, the regulatory capital required to cover losses at level α is

$$C_\alpha(\ell) = \|T\| \cdot \mathrm{VaR}_\alpha(\ell),$$

where $\mathrm{VaR}_\alpha(\ell)$ is the Value at Risk of ℓ under the induced probability $\mathbb{P}$.

In the above definition, the scaling by $\|T\|$ converts the quantile based risk into a balance value. This allows us to state and prove the following:

Theorem 4.2 (Esscher Premium as Regulatory Capital) *Let (γ, λ, T) be an accounting space and let ℓ be a loss attribute satisfying the conditions of Theorem 4.1 at confidence level α. Let $\theta^* = \delta_\alpha(\ell)$ be the corresponding PELV. Then the total Esscher premium for the loss attribute,*

$$\Pi_{\theta^*}(\ell) = \|T\| \cdot \mathbb{E}_{\mathbb{P}_{\theta^*}}[\ell],$$

exactly equals the regulatory capital requirement:

$$\Pi_{\theta^*}(\ell) = C_\alpha(\ell).$$

Proof By Theorem 4.1, θ^* uniquely satisfies $\mathbb{E}_{\mathbb{P}_{\theta^*}}[\ell] = \mathrm{VaR}_\alpha(\ell)$. Multiplying both sides by $\|T\|$ yields

$$\|T\| \, \mathbb{E}_{\mathbb{P}_{\theta^*}}[\ell] = \|T\| \, \mathrm{VaR}_\alpha(\ell) = C_\alpha(\ell)$$

□

Remark 11 Theorem 4.2 provides a rigorous economic interpretation of the Esscher loading parameter, that is θ^* is the degree of risk aversion implicit in the regulatory capital standard. The equality $\Pi_{\theta^*}(\ell) = C_\alpha(\ell)$ ensures that the risk-adjusted premium across all accounts precisely covers the capital required for extreme losses.

Example 9 (Insurance Pricing) Consider an insurer writing a portfolio of policies. Let $\ell(x)$ be the expected loss from policy x. Under Solvency II, the insurer must hold capital equal to the 99.5% VaR of its loss distribution. The PELV $\delta_{0.995}(\ell)$ gives the loading parameter θ^* such that the Esscher premium $\Pi_{\theta^*}(\ell)$ equals this capital requirement. The insurer can then set premiums based on $\Pi_{\theta^*}(\ell)$, ensuring that total premiums collected cover the regulatory capital charge.

Example 10 (Bank Loan Pricing) A bank's loan portfolio has loss attribute $\ell(x) = \mathrm{EAD}(x) \times \mathrm{PD}(x) \times \mathrm{LGD}(x)$. Under Basel III, the bank must hold capital against unexpected losses at the 99.9% confidence level. The PELV $\delta_{0.999}(\ell)$ yields the loading parameter that makes the Esscher premium equal to this capital charge. The bank can then allocate this capital charge to individual loans, pricing them according to their contribution to $\Pi_{\theta^*}(\ell)$ [10].

The above examples demonstrate that our rigorous mathematical model naturally connects existing methods in risk management to accounting structures, providing valid interpretations for accounting; in essence, this bridges accounting data, risk measurement, and regulatory policy, offering a unified framework for pricing, stress testing and capital allocation.

5 Temporal Accounting Space

In previous sections, we mainly developed the theory on static accounting spaces which capture the state of an entity at a single point in time. However, accounting is inherently dynamic: transactions occur, balances change, and financial statements are prepared over intervals, therefore a complete mathematical theory of accounting must be able to account for time. In this section, we extend the static framework to incorporate time, to what we call a *temporal accounting space*.

5.1 Temporal Extension

Let $\mathcal{T} = [0, \tau] \subset \mathbb{R}$ be a time interval with $\tau > 0$. Since we wish to model the balances with time, our temporal structure also needs to justify the changing balances over time. However, in practice, balances change only at discrete transaction moments and remain constant between them, and only at the exact instant of a transaction, the new balance is recorded immediately. This motivates requiring the balance paths to be càdlàg (right-continuous with left limits): at any transaction time t, the post-transaction balance is T_t, and the pre-transaction balance is the left limit T_{t-}.

Definition 5.1 (Temporal Accounting Space) A temporal accounting space over $\mathcal{T}$ is a family $\mathbb{F} = \{(\gamma, \lambda, T_t)\}_{t \in \mathcal{T}}$ such that:

1. For each $t \in \mathcal{T}$, (γ, λ, T_t) is an accounting space.
2. For every measurable set $A \in \lambda$, the function $t \mapsto T_t(A)$ is càdlàg.
3. There exists a countable set $\mathcal{D} \subset \mathcal{T}$ of transaction times such that $t \mapsto T_t(A)$ is continuous on $\mathcal{T} \setminus \mathcal{D}$.

Because each T_t individually satisfies the accounting equation $T_t(\gamma) = 0$, we have for all t:

$$T_t^+(\gamma) = T_t^-(\gamma) =: M_t, \qquad |T_t|(\gamma) = 2M_t =: ||T_t||,$$

where $||T_t|| > 0$ is assumed throughout (the entity has non-zero total absolute value at all times).

Since we have established a temporal accounting space, we may now introduce the idea of a transaction. Informally, a transaction is an event that redistributes value among accounts, which leads to a jump in the balance T_t of a particular account. We may note that the value of a transaction is that specific jump in balance, which we may formalise as the following:

Definition 5.2 (Transaction) Let $\mathbb{F} = \{(\gamma, \lambda, T_t)\}_{t \in \mathcal{T}}$ be a temporal accounting space. Denote the left limit $T_{t-}(A) = \lim_{s \uparrow t} T_s(A)$ from the càdlàg property, and the change of balance for a subaccount $A \subset \gamma$ by$\Delta T_t(A) = T_t(A) - T_{t-}(A)$. Then a transaction at time $t \in \mathcal{D}$ is a signed measure $\tau_t : \lambda \to \mathbb{R}$ defined by

$$\tau_t(A) = \Delta T_t(A)$$

for all $A \in \lambda$. Moreover it satisfies the following properties:

1. $\tau_t(\gamma) = 0$ (conservation of total balance),
2. countable additivity (the transaction distributes value across accounts),
3. $|\tau_t|(\gamma) < \infty$ (finite transaction size).

Note that the condition $\tau_t(\gamma) = 0$ is precisely the double-entry principle, that is, every debit is matched by a credit. Transactions may affect the total absolute value $||T_t||$ if they are related to incomes or losses, but they always preserve the net zero balance.

With the temporal accounting spaces, it is now possible to capture information about the account balances across time through integration. Now, when integrating over both time and accounts, the ability to interchange the order of integration is essential. For Fubini's theorem to apply, we need the family $\{|T_t|\}_{t \in \mathcal{T}}$ to be jointly measurable in (t, x); a sufficient condition for this is for the map $(t, x) \mapsto |T_t|(dx)$ to admit a measurable density with respect to a product reference measure, which will hold as long as each $|T_t|$ has a density $h(t, x)$ that is jointly measurable in (t, x) and integrable. This condition is relatively mild and is satisfied for the discrete case, where γ is countable, and continuous cases with piecewise continuous densities.

Theorem 5.1 (Fubini for Temporal Accounting Spaces) *Let $\mathbb{F} = \{(\gamma, \lambda, T_t)\}_{t \in \mathcal{T}}$ be a temporal accounting space such that the map $(t, x) \mapsto |T_t|(dx)$ is jointly measurable. Then for*

any non-negative measurable function $g : \mathcal{T} \times \gamma \to \mathbb{R}$*,*

$$\int_{\mathcal{T}} \int_{\gamma} g(t,x)\, |T_t|(dx)\, dt = \int_{\gamma} \int_{\mathcal{T}} g(t,x)\, dt\, |T_t|(dx).$$

The same holds for any g *integrable with respect to the product measure.*

Proof Since $|T_t|$ is a positive measure for each t and the map is jointly measurable, the function $(t,x) \mapsto g(t,x)$ is measurable with respect to the product σ-algebra. The result follows from the Fubini–Tonelli theorem applied to the product measure $\nu \otimes \mu$ where ν is Lebesgue measure on $\mathcal{T}$ and μ is defined by $\mu(A \times B) = \int_A |T_t|(B)\, dt$ for measurable $A \subset \mathcal{T}$, $B \in \lambda$. The extension to signed measures follows by decomposing into positive and negative parts. □

Example 11 (Cumulative Interest Earned) Suppose a bank earns interest daily on its loan portfolio. Let $r_t(x)$ be the interest rate applicable to loan x at time t, and let $|T_t|(\{x\})$ be the outstanding balance of loan x at time t. The total interest earned over a period $[0, \tau]$ is given by

$$I = \int_0^{\tau} \int_{\gamma} r_t(x)\, |T_t|(dx)\, dt.$$

Assume the family $\{|T_t|\}$ is jointly measurable so that Fubini's theorem applies. Define the measure μ on γ by

$$\mu(B) = \int_0^{\tau} |T_t|(B)\, dt, \qquad B \in \lambda.$$

Then μ represents the time-integrated absolute balance: for any loan x, $\mu(\{x\}) = \int_0^{\tau} |T_t|(\{x\})\, dt$ is the total dollar-days outstanding for that loan. By Fubini's theorem,

$$I = \int_{\gamma} \left(\int_0^{\tau} r_t(x)\, dt \right) \mu(dx).$$

Thus the total interest equals the sum over loans of the cumulative interest rate applied to each loan multiplied by the loan's time-integrated balance. In practice, if the interest rate is constant over the period, say $r_t(x) = r(x)$, then

$$I = \int_{\gamma} r(x) \left(\int_0^{\tau} |T_t|(dx)\, dt \right) = \int_{\gamma} r(x)\, \mu(dx),$$

which is simply the weighted sum of loan balances integrated over time.

5.2 Induced Probability from Temporal Accounting Spaces

Since we have constructed probability spaces in the static accounting spaces, we may naturally introduce induced probability spaces to the temporal accounting spaces. In particular at each time t, provided $||T_t|| > 0$, we have an induced probability measure

$$\mathbb{P}_t(E) = \frac{|T_t|(E)}{||T_t||}, \qquad E \in \lambda.$$

From the family $\{\mathbb{P}_t\}$, we can construct aggregated probability measures that summarise the entire period:

Definition 5.3 (Temporal Mixture Measures) For a temporal accounting space $\mathbb{F}$ over $[0, \tau]$, define:

1. The time-averaged probability measure

$$\bar{\mathbb{P}}(E) = \frac{1}{\tau} \int_0^\tau \mathbb{P}_t(E)\, dt, \qquad E \in \lambda.$$

2. The value-weighted temporal probability measure

$$\tilde{\mathbb{P}}(E) = \frac{\int_0^\tau ||T_t||\, \mathbb{P}_t(E)\, dt}{\int_0^\tau ||T_t||\, dt}, \qquad E \in \lambda.$$

The measure $\bar{\mathbb{P}}$ gives equal weight to each instant, while $\tilde{\mathbb{P}}$ weighs each time by the total size of the entity. The latter has a natural interpretation as the distribution of a randomly selected "dollar-day" – that is, pick a time t with probability proportional to $||T_t||$, then pick a balance unit of absolute value at that time according to $\mathbb{P}_t$.

The temporal mixture measures inherit many properties from the static case:

Proposition 5.2 *Let $\bar{\mathbb{P}}$ and $\tilde{\mathbb{P}}$ be defined as above. Then:*

1. *Both $\bar{\mathbb{P}}$ and $\tilde{\mathbb{P}}$ are probability measures on (γ, λ).*
2. *If the family $\{\mathbb{P}_t\}$ is stationary (i.e., $\mathbb{P}_t = \mathbb{P}_0$ for all t), then $\bar{\mathbb{P}} = \tilde{\mathbb{P}} = \mathbb{P}_0$.*
3. *For any bounded measurable attribute f,*

$$\mathbb{E}_{\bar{\mathbb{P}}}[f] = \frac{1}{\tau} \int_0^\tau \mathbb{E}_{\mathbb{P}_t}[f]\, dt,$$

$$\mathbb{E}_{\tilde{\mathbb{P}}}[f] = \frac{\int_0^\tau ||T_t||\, \mathbb{E}_{\mathbb{P}_t}[f]\, dt}{\int_0^\tau ||T_t||\, dt}.$$

4. *By Fubini's theorem,*

$$\mathbb{E}_{\tilde{\mathbb{P}}}[f] = \frac{1}{\int_0^\tau ||T_t||\, dt} \int_\gamma f(x) \left(\int_0^\tau |T_t|(dx)\, dt \right),$$

which expresses the value-weighted temporal expectation as an average over accounts weighted by their time-integrated absolute value.

Proof Properties (1)–(3) follow directly from the definitions and linearity of integration. Property (4) uses Fubini's theorem to exchange the order of integration:

$$\int_0^\tau ||T_t||\, \mathbb{E}_{\mathbb{P}_t}[f]\, dt = \int_0^\tau \int_\gamma f(x)\, |T_t|(dx)\, dt = \int_\gamma f(x) \left(\int_0^\tau |T_t|(dx)\, dt \right).$$

Dividing by $\int_0^\tau ||T_t||\, dt$ yields the result. □

We conclude this section by providing some examples of how one may use the induced probability spaces of temporal accounting spaces for accounting analysis.

Example 12 (Credit Card Portfolio Dynamics) Let (γ, λ, T_t) be the temporal accounting space of a credit card portfolio, where each elementary account x corresponds to a cardholder with balance $b_t(x)$ (positive for outstanding debt, negative for overpayments). For simplicity, assume no negative balances, so $T_t = T_t^+$. The total absolute value at time t is $||T_t|| = \sum_x b_t(x)$. The induced probability is $\mathbb{P}_t(\{x\}) = b_t(x)/||T_t||$.

Assume that we observe the portfolio over a quarter $\mathcal{T} = [0, 90]$, that is 90 days. Note that balances only change at transaction times, so suppose the following is the set containing transaction times, $\mathcal{D} = \{t_1, t_2, \ldots, t_m\}$ (for example, when purchases occur). Between transactions, $\mathbb{P}_t$ is constant. Now one would be interested in the average concentration over a quarter. Thus, we use the time-averaged probability measure $\bar{\mathbb{P}}$ (Definition 5.3), and moreover, the average Herfindahl index is

$$\overline{\mathrm{HHI}} = \sum_x \bar{\mathbb{P}}(\{x\})^2,$$

where $\bar{\mathbb{P}}(\{x\}) = \frac{1}{90}\int_0^{90} \mathbb{P}_t(\{x\})\, dt$. This can be computed directly from the transaction data.

Alternatively, the value-weighted temporal measure $\tilde{\mathbb{P}}$ gives the distribution of a randomly selected dollar-day. The corresponding average HHI is

$$\widetilde{\mathrm{HHI}} = \sum_x \tilde{\mathbb{P}}(\{x\})^2 = \frac{\int_0^{90} ||T_t||\, \mathrm{HHI}_t\ dt}{\int_0^{90} ||T_t||\, dt},$$

which weights periods with larger total balances more heavily. This measure is more sensitive to times when the portfolio is largest and concentration risk is highest.

Suppose a customer segment shows a rising $\mathbb{P}_t$ over the quarter. Using $\tilde{\mathbb{P}}$, one may compute the segment's value-weighted share $\tilde{\mathbb{P}}(\text{subprime})$ and compare it to the time-averaged share. If the value-weighted share is significantly larger, it indicates that the segment's exposure is concentrated in high-balance periods, signaling elevated risk.

Example 13 (Calculating CECL Lifetime Loss Rate) Under the Current Expected Credit Loss (CECL) accounting standard, banks must estimate lifetime expected losses for financial assets. We may formalise an approach to estimate this lifetime expected losses using temporal accounting spaces.

Let (γ, λ, T_t) be the temporal accounting space of a bank's loan portfolio over a period $[0, \tau]$. For each loan $x \in \gamma$, let $f(x)$ be its lifetime loss attribute (e.g., probability of default times loss given default, adjusted for prepayment). The balance measure T_t records outstanding principal at time t, with $T_t(\{x\})$ positive for loans. The total absolute value at time t is $||T_t|| = |T_t|(\gamma) = \sum_x |T_t(\{x\})|$. The induced probability at time t is $\mathbb{P}_t(\{x\}) = |T_t(\{x\})|/||T_t||$.

The total expected loss over the period is

$$L = \int_0^\tau \int_\gamma f(x)\, |T_t|(dx)\, dt.$$

By Fubini's theorem (Theorem 5.1),

$$L = \int_\gamma f(x) \left(\int_0^\tau |T_t|(dx)\, dt \right) = \int_\gamma f(x)\, \mu(dx),$$

where $\mu(B) = \int_0^\tau |T_t|(B)\,dt$ is the time-integrated absolute balance. The value-weighted temporal probability $\tilde{\mathbb{P}}$ (Definition 5.3) satisfies

$$\mathbb{E}_{\tilde{\mathbb{P}}}[f] = \frac{L}{\int_0^\tau ||T_t||\,dt}.$$

Thus the average loss rate under $\tilde{\mathbb{P}}$ multiplied by total dollar-days $\int_0^\tau ||T_t||\,dt$ gives the total expected loss.

6 Conclusion and Future Work

This paper has developed a rigorous measure-theoretic foundation for accounting by introducing the concept of an accounting space: a measurable space equipped with a signed measure representing balances; the construction of this space came about by first constructing accounts using elementary accounts via nested intervals, establishing that any accounts can be decomposed into arbitrarily fine components. Within this framework, fundamental accounting principles such as the accounting equation arise not as assumptions but as theorems derived from the Hahn and Jordan decomposition theorems. The construction demonstrates that double-entry accounting possesses an inherent mathematical structure capable of supporting formal proofs and algebraic reasoning.

We further showed that every accounting space naturally induces a family of probability measures via the total variation measure $|T|$. We defined the standard induced probability $\mathbb{P}$, which provides grounds for value-weighted risk analysis of accounts. Following this, we also developed alternative measures: positive/negative, power-weighted, and exponentially tilted (Esscher), each of which is rightfully appropriate for different circumstances of concentration and risk analysis. We showed that power-weighted probabilities can be naturally connected to Rényi Entropies and provided an analysis with valid accounting interpretations. We also showed that defining the Esscher measure yields a result with a natural interpretation in accounting; when the loading parameter θ is chosen as the VaR-based Probability Equivalent Level (PELV) introduced by Wen and Zhang (2025), the total Esscher premium for a loss attribute ℓ exactly equals the regulatory capital requirement. This equality provides a precise economic interpretation: the loading parameter used in regulatory capital standards is the degree of risk aversion that makes the risk-weighted average loss coincide with the quantile-based capital charge. The result bridges accounting data, risk measurement, and regulatory policy within a unified framework.

The paper also explored temporal extensions, defining accounting spaces where balances evolve as càdlàg functions and transactions appear as jumps. Finally, the paper links induced probability spaces to temporal accounting spaces, with examples to demonstrate its usage for dynamic accounting analysis, whilst upholding the CECL standard.

The framework opens several research directions:

- Stochastic dynamics: developing a full stochastic theory where $\{T_t\}$ is a random process, in particular with connection to mathematical finance and dynamic risk measures.

- Information theory: deeper explorations of entropy measures and applied usage of information theoretic tools with relation to the accounting space framework.
- Esscher measures in accounting: further study of multivariate extensions and connections to utility theory.
- Empirical applications: calibration of power-weighted and Esscher measures to real-world data across different asset classes and regulatory regimes.
- Comparability of Accounting Structures: develop a metric space consisting of different accounting entities with a concrete metric, possibly constructed from the balance function, to compare accounting structures.

In summary, this research establishes a rigorous mathematical foundation for analytical accounting. The notion of an accounting space, together with its induced probability measures, provides a unified language for expressing accounting principles, quantifying risk, and connecting to regulatory standards.

Declarations

The author certifies that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

Appendix A Proof of Theorem 2.1

Proof Proof of (i). By definition, $I_n(k)$ is the closed interval between the two real numbers, and its endpoints lie within $[a, b]$ because we take max with a and min with b. Hence $I_n(k) \subseteq [a, b]$.

Proof of (2). Let $\delta_n = \frac{b-a}{2^n}$. Then $\delta_{n+1} = \frac{\delta_n}{2} \leq \delta_n$. For the lower endpoint,

$$L_n(k) = \max\{a,\ k - \delta_n\}, \quad L_{n+1}(k) = \max\{a,\ k - \delta_{n+1}\}.$$

Since $k - \delta_{n+1} \geq k - \delta_n$ (because subtracting a smaller number), we have $L_{n+1}(k) \geq L_n(k)$. For the upper endpoint,

$$U_n(k) = \min\{b,\ k + \delta_n\}, \quad U_{n+1}(k) = \min\{b,\ k + \delta_{n+1}\}.$$

Because $k + \delta_{n+1} \leq k + \delta_n$, we obtain $U_{n+1}(k) \leq U_n(k)$. Therefore $I_{n+1}(k) \subseteq I_n(k)$.

Proof of (3). First, k belongs to every $I_n(k)$: indeed $a \leq k \leq b$ and $k - \delta_n \leq k \leq k + \delta_n$, so k satisfies both inequalities after taking max and min. Hence $k \in \bigcap_n I_n(k)$.

Now let x be any point in the intersection. Then for every n,

$$\max\{a, k - \delta_n\} \leq x \leq \min\{b, k + \delta_n\}.$$

These inequalities imply

$$k - \delta_n \leq x \leq k + \delta_n \quad \text{for all } n,$$

because x is at least the max, which is at least $k - \delta_n$, and at most the min, which is at most $k + \delta_n$. Consequently,

$$|x - k| \leq \delta_n \quad \text{for all } n.$$

Since $\delta_n = \frac{b-a}{2^n} \to 0$ as $n \to \infty$, we must have $|x - k| = 0$, i.e., $x = k$. Thus the intersection contains only k.

Proof of (4). If two points k and k' give the same sequence of intervals, then their intersections coincide. But we have just shown that the intersection of the sequence for k is $\{k\}$ and for k' is $\{k'\}$; therefore $k = k'$. Hence the map $k \mapsto (I_n(k))_n$ is injective, and by definition it is surjective onto its image. □

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